Project Report: Olist E-Commerce Control Tower: An End-to-End Supply Chain Analytics & Forecasting Engine

Author: Abhishek Kapoor

1. Executive Summary: The Business Problem & Proposed Solution

1.1. The Business Problem:
Olist, a rapidly growing Brazilian e-commerce marketplace, faced critical operational challenges hindering its scalability and profitability. Key issues included:

- **Inventory Mismatch:** Frequent stock-outs on popular products led to lost sales, while over-stocking of other items tied up capital and strained warehouse capacity.
- **Inconsistent Customer Experience:** Unpredictable delivery times and unreliable vendor performance were damaging the Olist brand reputation and leading to customer dissatisfaction.
- **Operational Inefficiency:** A lack of centralized visibility into the supply chain made it difficult to identify the root causes of logistics delays, poor vendor performance, and product quality issues.

1.2. Proposed Solution: The "Control Tower"
To address these challenges, I developed a comprehensive, data-driven "Supply Chain Control Tower." This project is an end-to-end analytical solution that integrates three core business pillars: **Logistics Optimization, Vendor Performance, and Demand Forecasting.** The final deliverable is a prototype for an automated system that provides actionable insights to key stakeholders, enabling a shift from reactive problem-solving to proactive, data-informed decision-making.

1.3. Key Technologies & Methodologies Used:

- **Data Engineering & Analysis:** Python (Pandas, NumPy) for processing and merging 8 relational datasets (~100k orders).
- **Visualization:** Matplotlib & Seaborn for static analysis; a prototype dashboard concept designed for Streamlit.
- **Time Series Forecasting:** Facebook Prophet for its robustness, interpretability, and ability to model multiple seasonalities.
- **Natural Language Processing (NLP):** Scikit-learn (CountVectorizer, LatentDirichletAllocation) for extracting insights from over 10,000 text reviews..

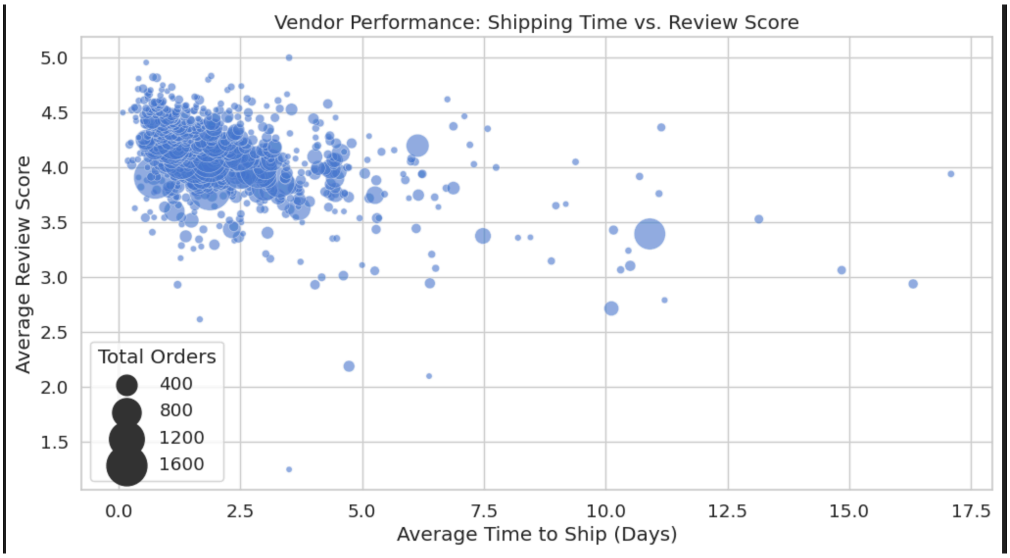
2. Exploratory Data Analysis (EDA): Uncovering Core Business Insights

The initial phase focused on deep exploration of the historical data (2017-2018) to establish a baseline understanding and answer immediate stakeholder questions.

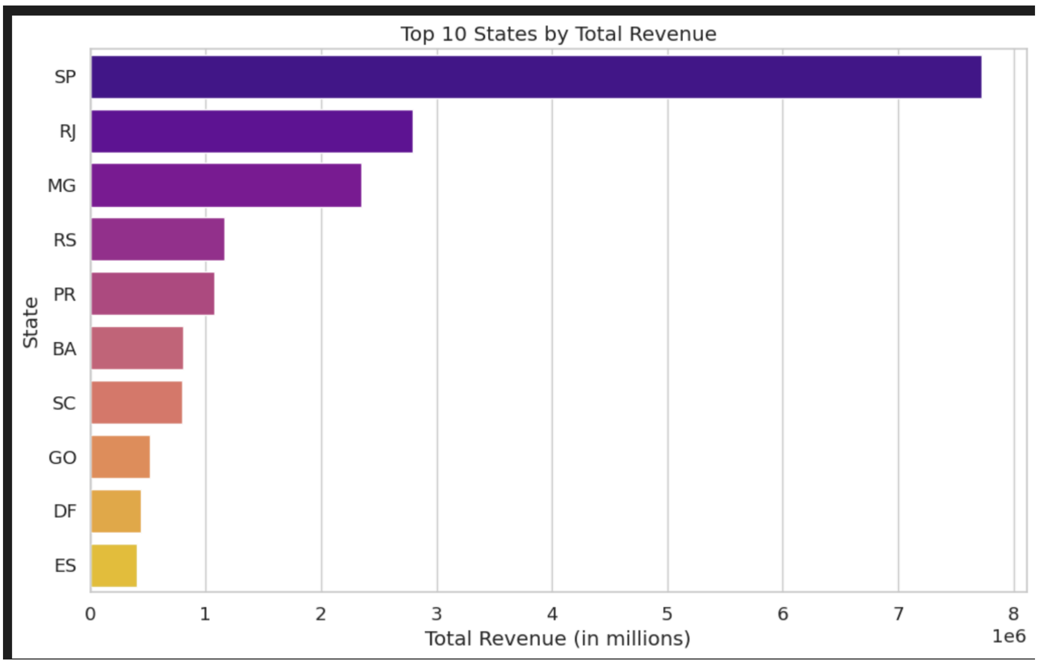
2.1. Logistics Performance Insights:

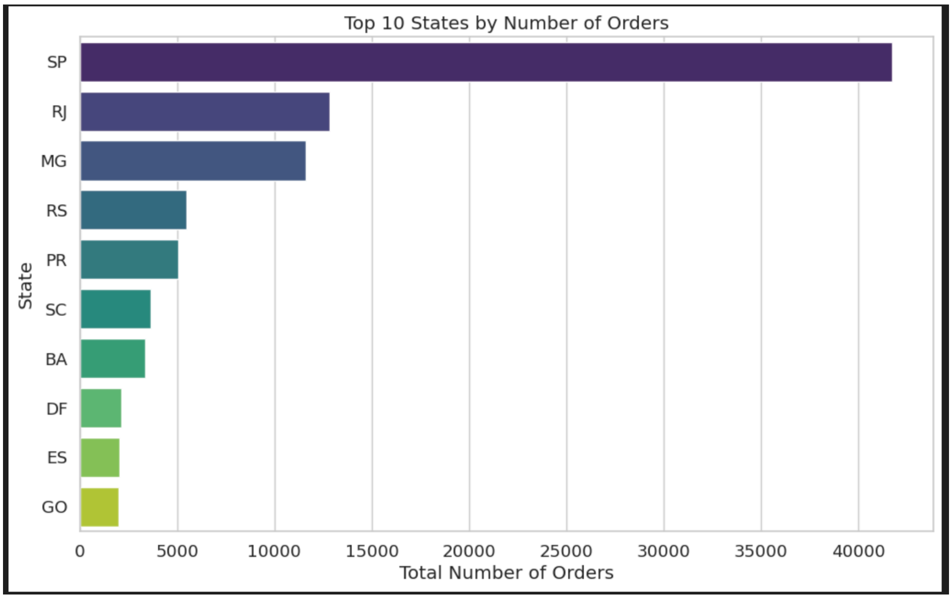
- **The "Last Mile" is the Longest Mile:** Analysis of delivery timelines revealed that, on average, a package spends **9.7 days in transit with the carrier**, compared to just **2.4 days being processed by the seller**. This crucial insight directs optimization efforts towards the carrier network, not just seller performance.

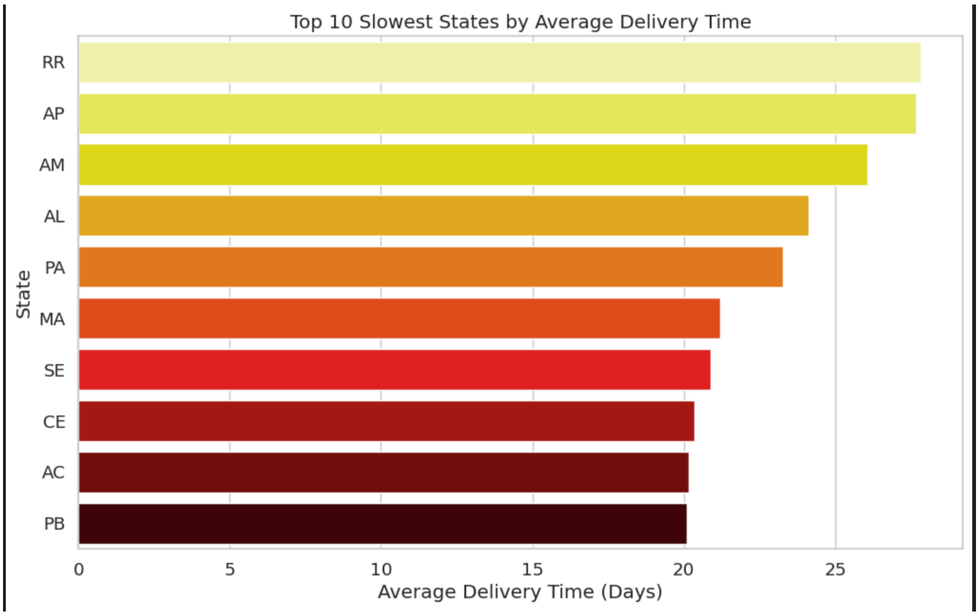
```
Average time from approval to shipment (Seller's responsibility): 2.41 days
Average time in transit (Carrier's responsibility): 9.67 days
On average, the seller takes 0.25 times longer than the carrier.
```



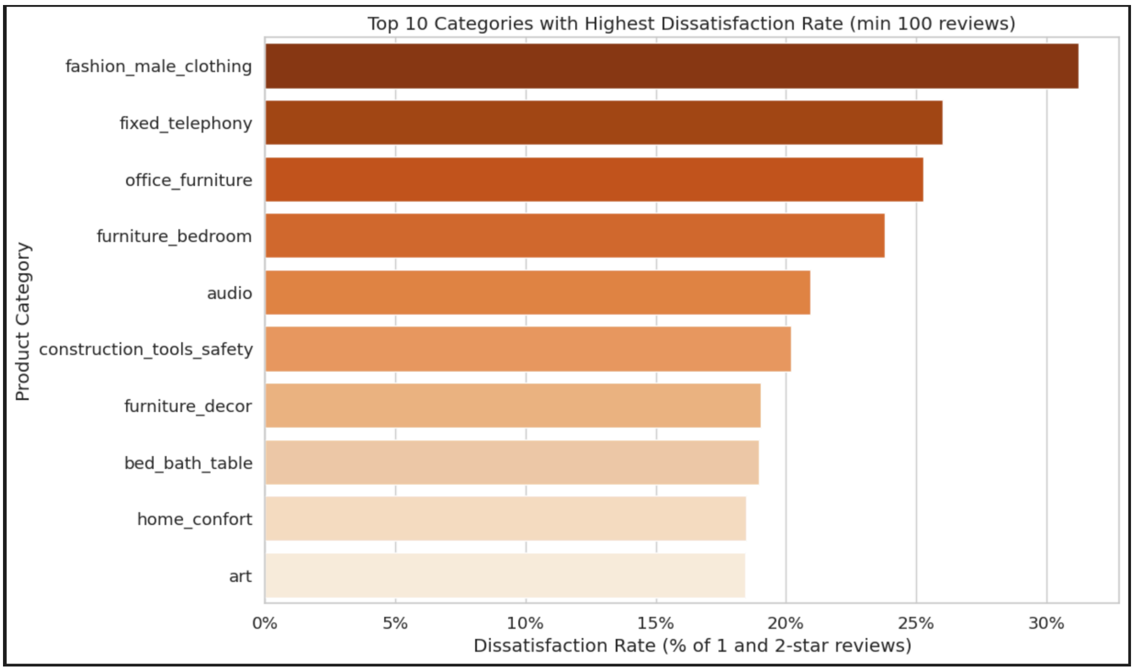
- **Geographical Disparity:** A clear performance gap exists between Brazil's regions. The home market of **São Paulo (SP)** boasts an average delivery time of ~8 days, while remote northern states like **Roraima (RR)** suffer from average delivery times exceeding 25 days.

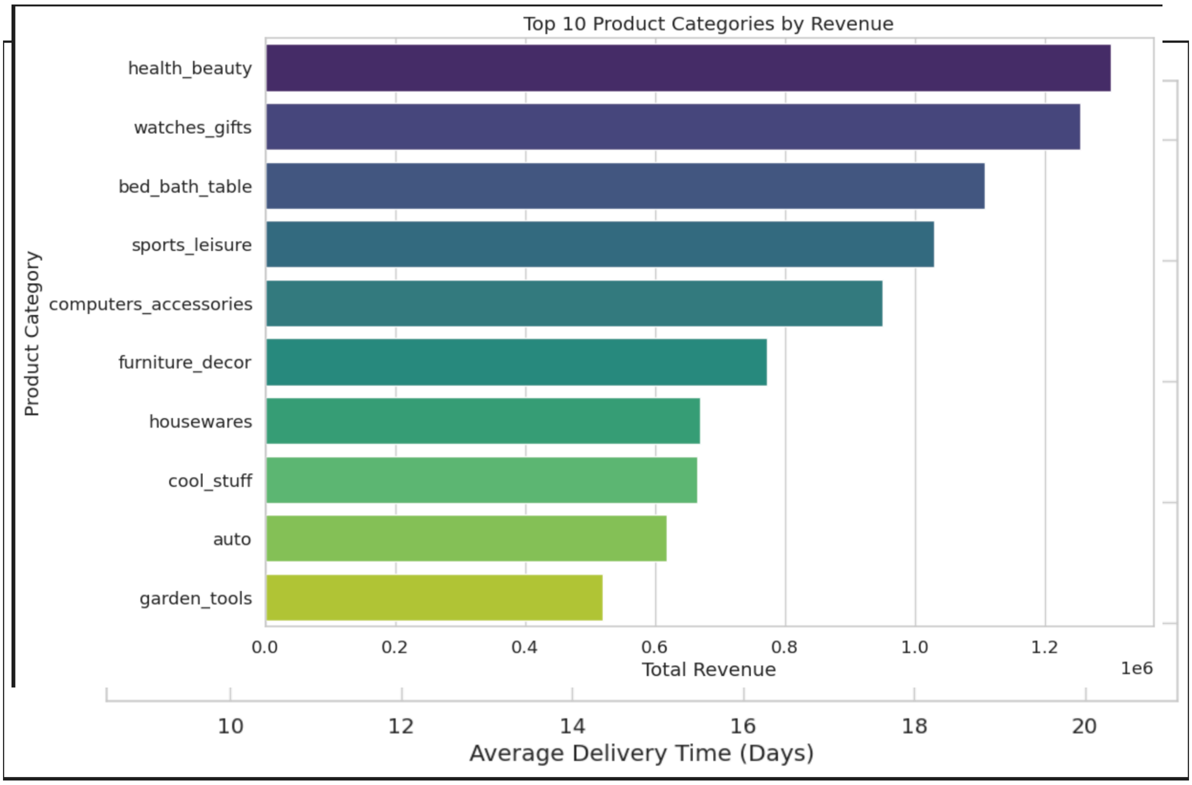


- 



- **Customer Expectations Management:** A positive finding was that **over 90% of orders are delivered on or before the estimated delivery date.** This indicates a successful strategy of setting conservative estimates to enhance customer satisfaction.



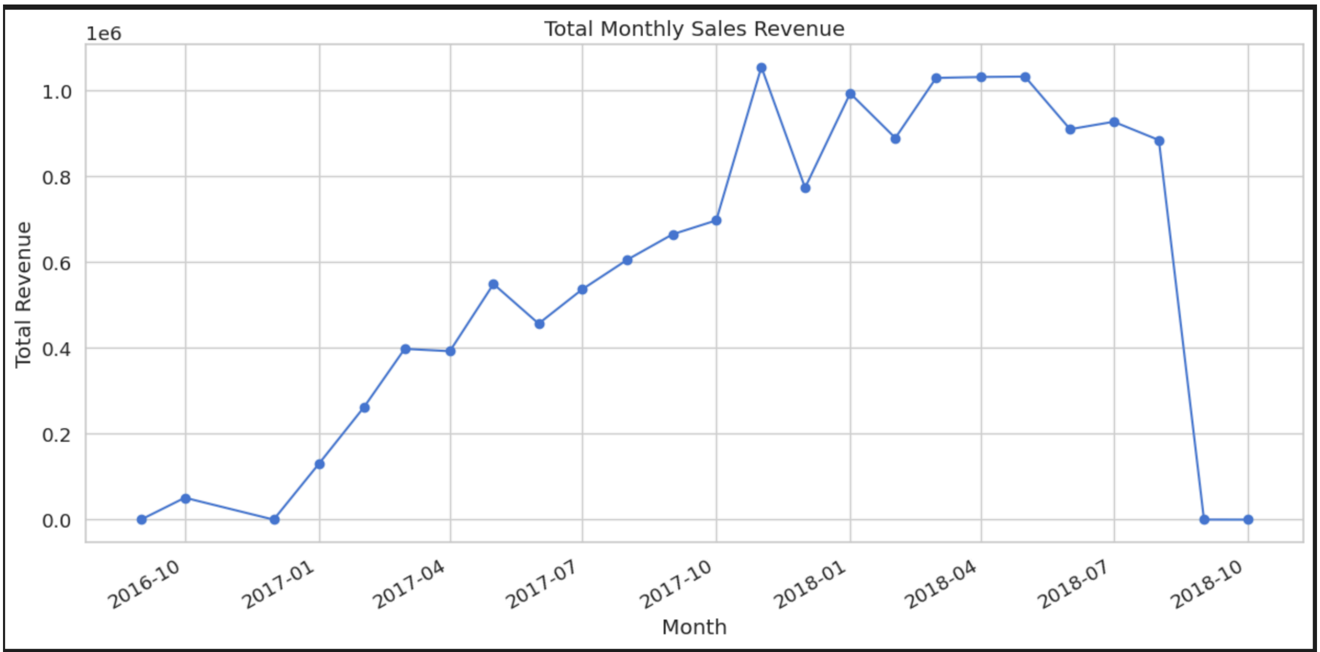


2.2. Vendor Performance Insights:

- **Speed is Quality:** A strong negative correlation was discovered between a vendor's average shipping time and their average customer review score. **Vendors who ship products faster consistently receive better ratings.**
- **High-Impact Vendors:** A vendor scoring system was created, combining review scores, shipping times, and order volume. This allowed for the clear identification of a small group of high-volume, low-performing vendors who represent a significant risk to Olist's reputation.

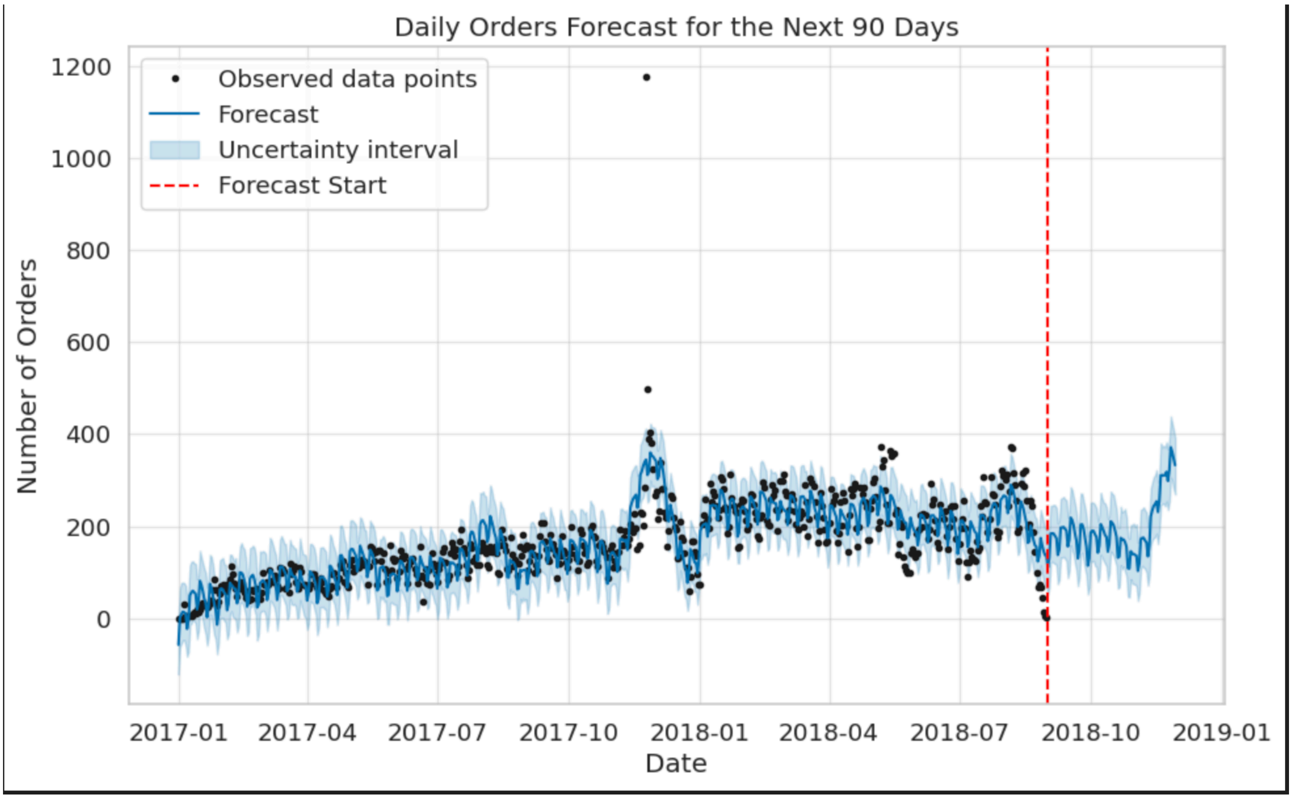
2.3. Demand & Product Insights:

- **Key Revenue Drivers:** The top 3 product categories—**Health & Beauty, Watches & Gifts, and Bed, Bath & Table**—were identified as the primary revenue drivers, highlighting where inventory management should be most stringent.



- **Sales Seasonality:** A massive, recurring sales spike was identified in late November, confirming the critical importance of the **Black Friday** period to Olist's annual revenue.

Of course. This is the perfect way to conclude the project. A clear, well-articulated comparison of



the modeling approaches, framed for both technical and business audiences, is the hallmark of a senior-level analyst or data scientist.

Here is the new, comprehensive version of the modeling and evaluation sections for your final project report.

3. Predictive Modeling: A Dual-Approach to Demand Forecasting

To address the critical business need for accurate inventory planning, a dual-modeling strategy was employed. This approach allowed for a robust comparison between a highly interpretable, specialized time-series model (Facebook Prophet) and a high-performance, feature-engineered machine learning model (LightGBM). The goal was not just to find the most accurate model, but to understand the trade-offs between interpretability, performance, and implementation complexity.

3.1. Baseline Model: Facebook Prophet for Interpretability

The initial forecasting model was built using Facebook Prophet. This tool was strategically chosen as the baseline for several key reasons:

- **For the Business User:** Prophet's primary advantage is its "glass-box" nature. It automatically decomposes the forecast into three intuitive components: the overall business **trend**, **weekly seasonality** (e.g., sales are lower on weekends), and **yearly seasonality** (e.g., the Black Friday spike). This is invaluable for building trust with the inventory team and explaining *why* the forecast looks the way it does, rather than just providing a number.